

architecture.md

docs | Markdown

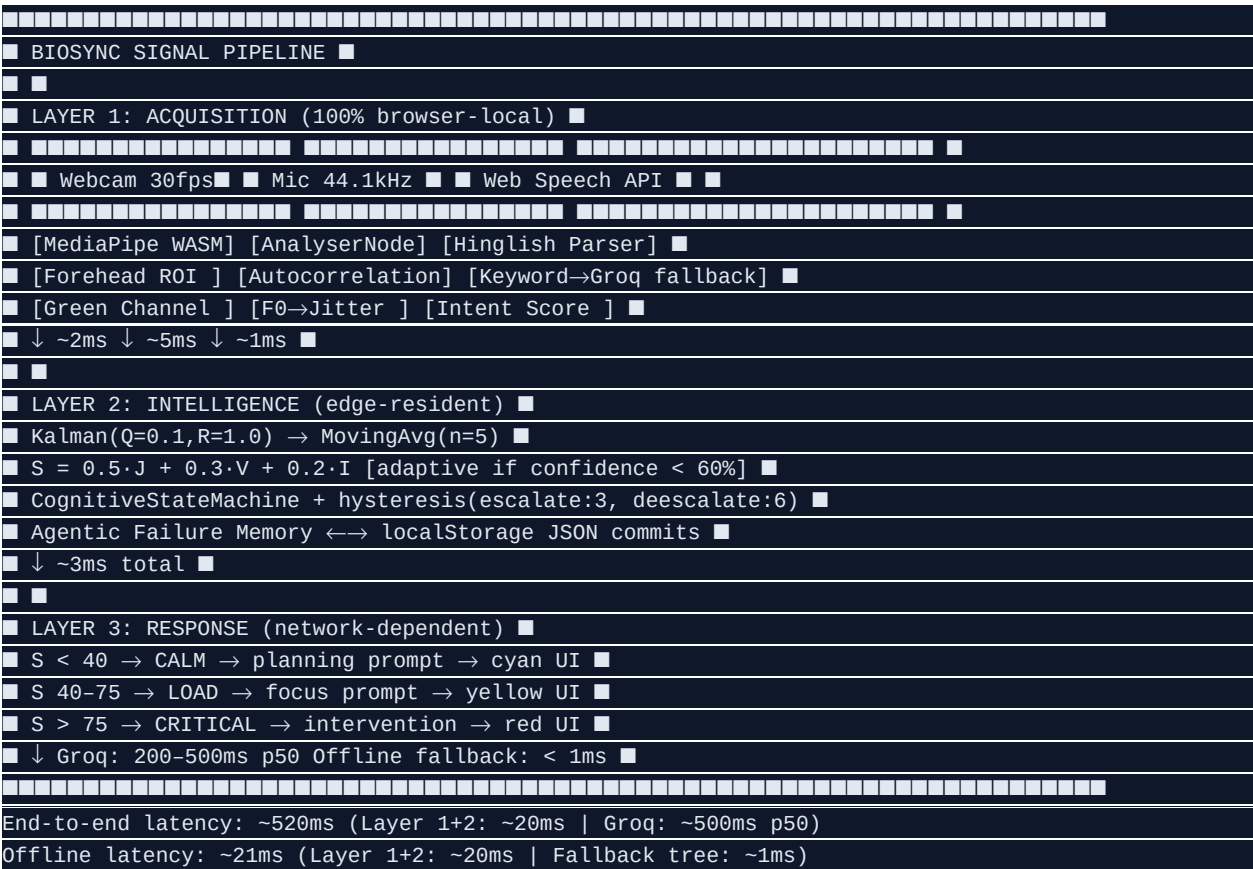
BioSync — System Architecture

Overview

BioSync is structured as three concentric layers with explicit responsibility boundaries.

The design constraint is edge-first: every computation that can run locally, does.

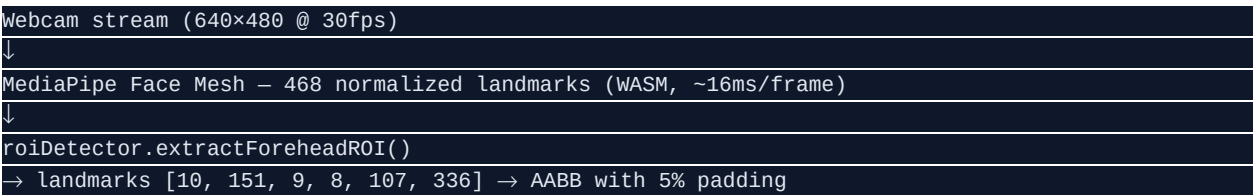
Only the AI response layer touches the network — and even that has a deterministic offline fallback.



Layer 1: Signal Acquisition

1.1 Visual Channel — rPPG

Files: signal-processing/rppgExtractor.ts, `signal-processing/roiDetector.ts`, `hooks/useRPPG.ts`



```

→ if area < 400px² → retry with extended landmark set [+54, 284, 103, 332]
→ isHeadPoseAcceptable() — yaw proxy via L/R eye z-depth differential
↓
Canvas getImageData(roi) → green channel mean:  $G_{\text{mean}} = \Sigma(\text{pixels}[i+1]) / n$ 
↓
KalmanFilter.update( $G_{\text{mean}}$ ) [Q=0.1, R=1.0]
↓
movingAverage(buffer, 5)
↓
computeNormalizedVariance(buffer, baseline, 30)
→  $V = \sigma^2(G_{\text{mean}}[t-30:t]) / \mu(G_{\text{mean}}_{\text{baseline}}) \times 1000$ , clamped 0–100

```

Why variance, not heart rate: BioSync uses HRV proxy — short-term variance in the rPPG signal — not absolute heart rate. Under sympathetic nervous system (SNS) activation, HRV decreases and becomes more erratic. This variance increase is measurable in ~1 second windows, whereas reliable HR extraction requires ≥ 5 seconds of clean signal. Variance-based detection is faster and more appropriate for real-time stress monitoring.

Why delta tracking beats absolute values: Raw green channel intensity varies by $\pm 30\%$ between users due to skin tone, lighting, and webcam response curves. Normalizing against a personal baseline (established in the 3-second calibration phase) makes the metric lighting-independent and skin-tone-independent — removing two major sources of demographic bias.

1.2 Audio Channel — Vocal Jitter

File: hooks/useVocalJitter.ts

```

Microphone stream (44.1kHz, mono)
↓
AudioContext → MediaStreamSource → AnalyserNode (FFT size: 2048, ~46ms window)
↓
getFloatTimeDomainData() every 500ms
↓
Voice activity detection: RMS > 0.01 threshold
↓ (if voiced)
Autocorrelation pitch detection:
lag ∈ [floor(44100/300), floor(44100/80)] → F0 in [80, 300] Hz
→ reject if normalized correlation < 0.01
↓
pitchHistory: rolling window of last 20 F0 values
↓
jitter =  $\Sigma |T(i+1) - T(i)| / (N-1) / T_{\text{mean}}$  ( $T = 1/F0$ , period in seconds)
↓
Normalize:  $((\text{jitter} - 0.8\%) / (4.0\% - 0.8\%)) \times 100$ , clamped 0–100

```

Physiological pathway: Psychological stress → HPA axis activation → cortisol release → laryngeal muscle tension increase → pitch period instability → elevated jitter.
Normal range: 0.5–1.0%. Stress-elevated: >1.5% (Goberman & Elmer, 2003).

Why audio weight is highest (0.5): Jitter provides a direct cortisol proxy via a physiological pathway completely independent of the visual channel. When both channels elevate simultaneously,

the fusion model gains confidence. When only one is available (e.g., no speech), the other compensates.

1.3 Semantic Channel — Hinglish NLP

File: `utils/hinglishParser.ts`

```
Speech transcript / typed input
↓
detectLanguage() – Hindi word frequency ratio
→ hindiRatio ≥ 0.5 → 'hindi'
→ hindiRatio ≥ 0.15 → 'hinglish'
→ else → 'english'
↓
Pattern matching against PATTERNS[] (regex × weight × intent)
→ 4-tier urgency: URGENT_DEADLINE (0.95) > HIGH_STRESS (0.90) >
OVERWHELMED (0.85) > FOCUS_REQUEST (0.70)
↓
Select highest-weight match → { intent, score, normalized: score×100 }
↓
Fallback to Groq if no patterns match and input length > 5 words
```

Layer 2: Intelligence Engine

2.1 Signal Smoothing

File: `utils/kalmanFilter.ts`

Two-pass smoothing pipeline applied to the rPPG signal:

Pass 1 — Kalman Filter (primary):

Optimal recursive Bayesian estimator for linear systems with Gaussian noise.

```
Prediction:  $P(t) = P(t-1) + Q$ 
Update:  $K = P(t) / (P(t) + R)$ 
 $x(t) = x(t-1) + K \times (z(t) - x(t-1))$ 
 $P(t) = (1 - K) \times P(t)$ 
```

Parameters $Q=0.1$, $R=1.0$ mean: trust the model 10× more than individual measurements.

This is appropriate for slow-varying physiological signals at 30fps. For high-motion demo environments, increase R to 2.0–3.0 for more aggressive smoothing.

Pass 2 — Moving Average (secondary):

5-sample (167ms) simple moving average after Kalman to remove residual quantization noise.

Computationally $O(n)$ but applied to a short window — negligible cost.

2.2 Weighted Fusion Model

File: `utils/stressEngine.ts`

```
 $S = (J \times 0.5) + (V \times 0.3) + (I \times 0.2)$ 
```

Where J = vocal jitter score, V = rPPG variance score, I = semantic intent score (all 0–100).

Standard weights (signalConfidence ≥ 60%):

Channel	Weight	Rationale
Audio (J)	0.5	Direct cortisol proxy; orthogonal to visual; continuous when speech active
Visual (V)	0.3	Peer-validated technique; SNR varies with consumer hardware
Semantic (I)	0.2	Discrete, event-driven; tiebreaker only

Adaptive weights (rPPG signalConfidence < 60%):

When the rPPG channel degrades (poor lighting, occlusion, head movement), the system redistributes weights to maintain fusion stability:

$$S = (J \times 0.5) + (V \times 0.1) + (I \times 0.4)$$

Audio remains primary. Semantic gains weight to compensate for reduced visual signal.

The fusion engine never divides by zero — if all signals are zero, score remains 0 (CALM).

2.3 State Machine with Asymmetric Hysteresis

File: utils/stressEngine.ts — `CognitiveStateMachine` class

CALM ($S < 40$) ↔ LOAD ($S \text{ } 40\text{--}75$) ↔ CRITICAL ($S > 75$)

Asymmetric hold times:

ESCALATION: breach threshold for ≥ 3 consecutive windows (~1.5s) → transition up
DE-ESCALATION: below threshold for ≥ 6 consecutive windows (~3.0s) → transition down

The 3:6 ratio is not arbitrary. It mirrors the physiological asymmetry of the HPA axis stress response: onset is fast (seconds), recovery is slow (minutes). The implementation matches this biology: fast to detect stress, conservative about declaring recovery.

Reference: Dickerson & Kemeny (2004), Psychological Bulletin.

2.4 Agentic Failure Memory

File: hooks/useStateMemory.ts

Each AI intervention is stored as a typed `InterventionCommit` in `localStorage`:

```
{
  id, sessionId, timestamp,
  interventionType, // e.g. 'BREATHING_EXERCISE'
  interventionText, // exact AI response shown
  preScore, // fusion score at trigger
  postScore, // measured 60s post-intervention
  delta, // postScore - preScore
  outcome // 'SUCCESS' if delta ≤ -10, else 'FAILED'
}
```

Outcome threshold: $\text{delta} \leq -10$ (score drops by ≥10% of full scale). A smaller threshold would produce false successes; a larger threshold would be too conservative for a 60-second window. Both thresholds are exported as named constants and are tunable.

The AI prompt enforces: never suggest any intervention in this array.

```
→ jitterScore (0-100) ████████→ state: CALM / LOAD / CRITICAL
↓
[Speech / text] Agentic Failure Memory
→ detectLanguage ← read blacklist
→ PATTERNS matching → write outcome commits
→ semanticScore (0-100) ████████████████████████
↓
getBioSyncIntervention()
→ Groq (online) or fallback tree (offline)
↓
UI: color transition + waveform + score + text
```

Privacy Architecture

Data type	Where processed	Transmitted?	Stored?
Video frames	Canvas (browser)	Never	Never
Audio PCM	AudioContext (browser)	Never	Never
Speech text	Web Speech API (browser)	Never externally	Never
rPPG score	Browser memory	Groq receives score only	Session RAM
Jitter score	Browser memory	Groq receives score only	Session RAM
Intervention log	localStorage	Never	User-deletable

DPDP Act 2023 compliance: purpose-limited (cognitive load only), locally processed, no persistent biometric storage, user-deletable at any time.

Performance Characteristics

Operation	Latency	Complexity	Notes
MediaPipe face detection	~16ms	O(n landmarks)	WASM, 60fps capable
ROI extraction + pixel read	~2ms	O(roi pixels)	Canvas getImageData
Kalman filter update	< 0.1ms	O(1)	Constant-time recursive
Moving average	< 0.1ms	O(window)	window=5
Variance computation	< 1ms	O(window)	window=30
Jitter computation	~1ms	O(history)	history≤20 pitches
Hinglish pattern matching	< 0.5ms	O(patterns)	18 patterns
Fusion score + state machine	< 0.1ms	O(1)	Arithmetic only
Groq inference (online)	200-500ms	–	Network p50
Offline fallback tree	< 1ms	O(candidates)	Deterministic

Total online latency: ~520ms
Total offline latency (Layers 1+2 only): ~20ms